

5.23 (a) The cylinder can be viewed as a stack of current loops. From Eq. (5.103), the current loops have radius a and current intensity M in the azimuthal direction $\hat{\phi}$, parallel to the surface. Therefore, the image current will be identical to the equivalent currents.

Repeat Problem 5.18 (a), we can see that, for two coaxial circular loops with currents in the same direction and separated by a distance of d , the force between them is

$$F(d) = \mu_0 I^2 a \frac{\partial}{\partial d} \left\{ \left(\frac{2}{k(d)} - k(d) \right) K(k(d)) - \frac{2}{k(d)} E(k(d)) \right\},$$

where $k(d) = 2a / \sqrt{4a^2 + d^2}$. Then, the total force becomes

$$\begin{aligned} F &= \int_0^L dz_1 \int_0^L dz_2 F(z_1 + z_2) \\ &= \mu_0 I^2 a \int_0^L dz_1 \left\{ \left[\left(\frac{2}{k(z_1+L)} - k(z_1+L) \right) K(k(z_1+L)) - \frac{2}{k(z_1+L)} E(k(z_1+L)) \right] \right. \\ &\quad \left. - \left[\left(\frac{2}{k(z_1)} - k(z_1) \right) K(k(z_1)) - \frac{2}{k(z_1)} E(k(z_1)) \right] \right\} \end{aligned}$$

It is possible to show that

$$\int \left[\left(\frac{2}{k(z)} - k(z) \right) K(k(z)) - \frac{2}{k(z)} E(k(z)) \right] dz = \frac{K(k(z)) - E(k(z))}{k(z)} \cdot z + C,$$

then

$$\begin{aligned} F &= \mu_0 I^2 a \left[\frac{K(k(z_1+L)) - E(k(z_1+L))}{k(z_1+L)} \cdot (z_1+L) \Big|_0^L - \frac{K(k(z_1)) - E(k(z_1))}{k(z_1)} \cdot z_1 \Big|_0^L \right] \\ &= 2\mu_0 a L I^2 \left[\frac{K(k(2L)) - E(k(2L))}{k(2L)} - \frac{K(k(L)) - E(k(L))}{k(L)} \right] \end{aligned}$$

Let $k_1 = k(2L) = a / \sqrt{a^2 + L^2}$, $k = k(L) = 2a / \sqrt{4a^2 + L^2}$, and notice that $I = M$, we

will finally have

$$F = -2\mu_0 a L M^2 \left[\frac{K(k) - E(k)}{k} - \frac{K(k_1) - E(k_1)}{k_1} \right],$$

which differs Jackson by a minus sign. Here, the minus sign indicates attractive force.

(b) Using the MacLaurin expansion,

$$K(k) = \frac{\pi}{2} + \frac{\pi}{8} \frac{k^2}{1-k^2} + \dots, \quad E(k) = \frac{\pi}{2} - \frac{\pi}{8} k^2,$$

When $L \gg a$, k and k_1 are small, and

$$F = -2\mu_0 a L M^2 \cdot \left(\frac{\pi}{4} k - \frac{\pi}{4} k_1 \right) \rightarrow -2\mu_0 a L M^2 \cdot \frac{\pi}{4} \frac{a}{L} = -\frac{1}{2} \mu_0 \pi a^2 M^2,$$

Which agrees with Problem 5.22.